

Htet Myat Aung (Alexander E. Hong)

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Education

King Mongkut's University of Technology Thonburi (KMUTT) , Automation Engineering - Thesis: Natural Gas Methane Leak Detection Using an Infrared Camera - INC Department Outstanding Student Scholarship - Winter Semester 2023 & 2024	Bangkok, Thailand 2022 – 2025
Yangon Technological University (Center of Excellence) , Mechatronics Engineering - Participated in Myanmar RoboLeague 2019 as Mentor - Led First Global Challenge 2019 team representing Myanmar	Yangon, Myanmar 2015 – 2022

Experience

Maintenance Technology Center (MTC) , Full Time Machine Learning Researcher Developing advanced machine learning solutions for non-destructive testing applications. Collaborating with Prof. Cherdpong Jomdecha on cutting-edge detection technology research. - Contributing to innovative maintenance technology solutions for industrial applications - Research focus on non-destructive detection technology	Bangkok, Thailand Sept 2025 – present 5 months
Maintenance Technology Center (MTC) , Machine Learning Intern Developed supervised, unsupervised and semi-supervised machine learning algorithms for acoustic detection based non-destructive detection systems. - Won First Runner-Up Award in 'Smart Infrastructure for Smart City' competition by Metropolitan Waterworks Authority - Listed in KMUTT Hall of Fame for outstanding achievement in Smart Infrastructure development - Applied AI/ML techniques to solve real-world infrastructure challenges	Bangkok, Thailand June 2025 – Aug 2025 3 months
HoneyHunt.AI , Junior Machine Learning Intern (LLM Intern) Enhanced automated resume parsing and intelligent job matching systems utilizing advanced LLM APIs. - Improved algorithm scalability and matching accuracy for production-level deployment - Contributed to product development in AI-powered recruitment technology	Bangkok, Thailand Dec 2024 – Feb 2025 3 months

Research

KMUTT - OPTIMISED 3D-CNN FOR REALTIME INFRARED NATURAL GAS LEAK CLASSIFICATION , Student Researcher Developed a high-accuracy (99.75%) 3D CNN model for methane gas leak detection. Co-Winner of Best Paper Award at IcoICT 2025 (IEEE Xplore). - Applied Running Average, GMM, Moving Average, and Background Subtraction techniques - Implemented smaller image sizes achieving above 99% accuracy for efficient deployment - Currently developing quantization techniques with Asst. Prof. Dr. Sarawan Wongsra for journal publication	Bangkok, Thailand June 2024 – present
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CODIA Lab, KMUTT & Panasonic Singapore, Student Researcher - SAM and Computer Vision Systems Development
Developed SLAM Sparse Bundle Adjustment algorithms in C for edge devices.

- Enhanced algorithm efficiency for edge device deployment

Bangkok, Thailand
Feb 2024 – July 2024

COSIN Lab, INC Department, Student Researcher - Autonomous Vehicle Control Systems
Developed ROS2-based autonomous navigation and control systems for competitive robotics applications.

- Integrated micro-ROS framework with IMU sensor data for enhanced navigation precision
- Contributed to robotics competition preparation and system optimization

Bangkok, Thailand
Nov 2023 – Jan 2024

Volunteer

FIRST GLOBAL CHALLENGE 2019, Mentor
Provided technical guidance and strategic mentorship to Myanmar national robotics team.

- Led team to win prestigious Judges Award at international robotics competition
- Team placed 82 out of 189 countries
- Fostered STEM education and international collaboration in robotics

Dubai, UAE
June 2019 – Oct 2019

Laboratory of Robotics and Artificial Intelligence (L.O.R.A), Robotics Researcher and Founder
Founded and led comprehensive robotics research laboratory focusing on practical applications.

- Spearheaded development of autonomous disinfection robots during COVID-19 pandemic
- Successfully donated first disinfection robots to Wai Bar Gi Hospital
- Managed multiple concurrent research initiatives and mentored junior researchers
- Grew organization to 44 undergraduate student members

Yangon, Myanmar
July 2019 – June 2021

ConceptX International School, Head of Department and Instructor of Robotics and Computer Science
Instructed Computer Science and Robotics to approximately 200 students from Upper Primary 5 to Lower Secondary 2.

- Promoted to Head of Department in October 2021
- Taught IGCSE Computer Science to Secondary 3 students

Yangon, Myanmar
June 2021 – Dec 2021

Awards

Best Paper Award - IcoICT 2025 (IEEE Xplore): Co-winner with Asst. Prof. Dr. Sarawan Wongsu, 13th International Conference on Information and Communication Technology, Bandung

Best Presentation Award - ICA-SP Con 2025: 16th Instrumentation, Control and Automation Senior Project Conference

First Runner-Up - Smart Infrastructure for Smart City Competition: Metropolitan Waterworks Authority, Bangkok

KMUTT Hall of Fame: King Mongkut's University of Technology Thonburi

INC Department Outstanding Student Scholarship: KMUTT - Winter Semester 2024

INC Department Outstanding Student Scholarship: KMUTT - Winter Semester 2023

Judges Award - First Global Challenge 2019: International Robotics Competition, Dubai, UAE

Publications

Optimized 3D-CNN for Real-Time Infrared Natural Gas Leak Classification

Developed a high-accuracy (99.75%) 3D CNN model for methane gas leak detection using infrared imaging. Winner of Best Paper Award. DOI - 10.1109/ICoICT66265.2025.11192973
Htet Myat Aung, Sarawan Wongsu
ieeexplore.ieee.org/document/11192973

Natural Gas Methane Leak Detection Using an Infrared Camera

Best Presentation Award winner for research on natural gas leak detection systems using infrared imaging technology.
Htet Myat Aung

Skills

Programming Languages

AI/ML Frameworks

Robotics & Systems

Core Competencies

Languages

English

Fluent

Burmese (Myanmar)

Native speaker

Interests

Artificial Intelligence & Machine Learning

Robotics & Automation

Research & Innovation